Data Flow Diagram (DFD)

iMovies MERN Project • Phase-wise Documentation

Project: iMovies MERN **Document:** Data Flow Diagram (DFD)

1. Purpose and notation

This document describes the main data flows for iMovies. External entities are Users and Administrator; processes are application functions; data stores are MongoDB collections. The diagrams are logical and should be checked against the final implementation.

2. Context diagram

External entity	Flow	System
User	Registration/login, movie search, show and seat selection, booking requests	iMovies Web Application
iMovies Web Application	Movie details, auth result, booking status and booking history	User
Administrator	Admin login, movie create/delete, booking review	iMovies Web Application
iMovies Web Application	Admin status, movie/booking information	Administrator

3. Level 1 DFD

Process	Input	Data store / output
1. Authenticate user	Registration/login details	User collection; token/result returned.
2. Browse movies	Search/filter query	Movie collection; movie list returned.
3. Create booking	User identity, movie, show, seats	Movie and reservation checks; Booking and SeatReservation records; result returned.
4. Manage own bookings	Authenticated user ID, list/cancel request	Booking collection filtered by user; history/status returned.
5. Admin management	Admin token and movie/booking action	Movie and Booking collections; operation result returned.

4. Text flow diagram

USER / ADMIN
↓ requests through browser
REACT FRONTEND
↓ HTTP/JSON API request
EXPRESS + NODE.JS API
■■ Authentication / role validation
■■ Input validation and booking rules
↓ Mongoose operations
MONGODB (Users | Movies | Bookings | SeatReservations)
↑ query result / saved result
EXPRESS API → JSON response → REACT UI

5. Data descriptions

Data item	Origin	Use
Credentials	User form	Authenticate; passwords must be hashed at rest.
Movie details	Admin / Movie collection	Display movie catalogue.
Booking request	Signed-in user	Validate selected show and seats.
Booking record	Backend	Track owner, movie, seats, amount and status.
Admin action	Authorized administrator	Maintain catalogue and review bookings.

6. Data protection

Never place passwords, JWT secrets or database credentials in frontend code or public documentation. Use authorization checks on the server, and only return the data each role is allowed to access.

7. Validation note

The current diagrams are a logical representation. Confirm exact field names, endpoint names and reservation flow from server.js and the Mongoose models before treating this as an implementation-level DFD.

Prepared for academic project documentation. Verify implementation-specific details and fill in actual test evidence before final submission.